

TECHNICAL REPORT

SMART ADAPTIVE TRAFFIC SIGNAL SYSTEM

Reducing Waiting, Idling and Traffic Congestion at Intersections

Group Theme: **Traffic Congestion and Vehicle Emissions**

Sustainable Transportation Planner

Submitted by: JOVEEN JESS CHIRAYATH

Roll No.: **22**

U2503163

S3 CS Gamma

1. Abstract

Traffic congestion at signalized intersections causes significant delay, repeated stopping and vehicle idling. Conventional fixed-time traffic signals operate according to predetermined schedules and cannot always respond to rapidly changing traffic demand. The Smart Adaptive Traffic Signal System (SATSS) proposed in this report uses real-time traffic information from sensors or cameras to adjust signal timing dynamically within predefined safety limits. Traffic volume, queue length, waiting time and approach demand are processed to determine suitable green-time allocation. By responding to actual conditions, the system aims to reduce unnecessary waiting and idling, improve traffic flow and potentially lower fuel consumption and associated emissions. The system can be integrated with the Sustainable Transportation Planner website to display traffic conditions, signal status and performance indicators. The proposal combines transportation engineering, sensing, data processing and intelligent control to provide a practical sustainability-oriented solution.

2. Introduction

Urban traffic demand varies throughout the day because of commuting peaks, school hours, incidents, weather, road works and special events. A fixed signal plan may therefore give the same green-time allocation even when traffic demand is highly unequal between approaches. One approach may develop a long queue while another receives green time with little traffic.

Vehicles waiting at intersections spend time without making progress. When many vehicles queue simultaneously, the result is increased travel time, fuel use and transport-related emissions. Adaptive signal control attempts to use existing road capacity more efficiently by changing signal timings in response to current traffic demand.

The proposed system is designed as a feedback-control system. It observes traffic, processes the measurements, makes a timing decision subject to safety constraints, and then observes the new traffic condition. This allows the system to respond to demand while retaining minimum green, maximum green, yellow, all-red and pedestrian-safety requirements.

3. Problem Statement

The problem addressed by this project is inefficient traffic-signal operation when actual traffic demand differs from the fixed timing plan. Poor allocation of green time can create long queues, increase waiting time and cause unnecessary idling. A responsive system is needed to improve intersection efficiency without compromising safety.

3.1 Limitations of Fixed-Time Signals

- Predefined timings may not match real-time traffic conditions.
- Low-demand approaches may receive unnecessary green time.
- Heavy-demand approaches can develop long queues and excessive delay.
- Incidents and unusual traffic surges may not be handled quickly.
- Repeated stopping and idling can increase fuel use and emissions.

3.2 Objectives

1. Measure traffic volume, density and queue conditions at intersections.
2. Adjust green time according to current traffic demand.
3. Reduce average waiting time and unnecessary idling.
4. Improve traffic throughput and travel-time reliability.
5. Estimate environmental performance indicators.
6. Provide a dashboard for monitoring system performance.

3.3 Proposed Performance Measures

Measure	Purpose
Average delay	Measures waiting efficiency
Queue length	Indicates congestion
Number of stops	Indicates interruption of flow
Estimated idle time	Supports emission assessment
Throughput	Measures vehicles served

4. Sustainability Analysis

4.1 Relevant SDGs

The proposed system contributes most directly to SDG 11 – Sustainable Cities and Communities. Target 11.2 promotes safe, affordable, accessible and sustainable transport systems. The project also supports SDG 9 – Industry, Innovation and Infrastructure through intelligent transport infrastructure, and SDG 13 – Climate Action through the potential reduction of avoidable idling, fuel use and associated emissions.

SDG	Project Relevance
SDG 9	Smart and innovative transport infrastructure
SDG 11	More efficient and sustainable urban mobility
SDG 13	Potential reduction in idling-related emissions

4.2 ISO 14001 Perspective

ISO 14001 provides a framework for managing environmental aspects and improving environmental performance. For this project, the approach can be applied by identifying vehicle delay and emissions as relevant environmental aspects, setting measurable objectives, monitoring indicators and continuously improving system performance. Useful indicators include delay, queue length, estimated idle time, fuel-use proxy and estimated CO₂ emissions.

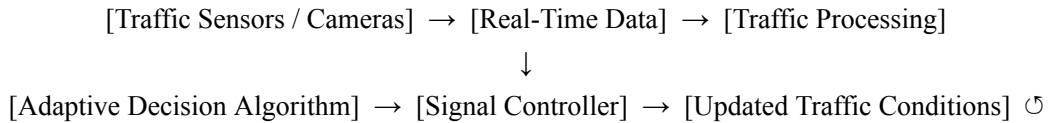
4.3 LCA/EIA Considerations

A complete Life Cycle Assessment (LCA) or Environmental Impact Assessment (EIA) requires site-specific data and is outside the scope of this student proposal. A qualitative assessment should nevertheless consider both benefits and impacts. Benefits may include reduced vehicle delay and emissions. Impacts may arise from manufacturing, electricity use, maintenance and disposal of sensors, cameras, controllers and communication equipment. Energy efficiency, equipment lifetime, repairability and responsible electronic-waste management should therefore be considered in a real deployment.

5. Proposed Engineering/IT Solution

The SATSS is a closed-loop control system that continuously observes traffic conditions and adjusts signal timing within predefined safety constraints. The main components are traffic detectors, a processing unit, an adaptive decision algorithm, a signal controller, a database and a web dashboard.

5.1 System Architecture



Sensors may measure vehicle count, occupancy, approximate speed and queue length. The processing unit filters and aggregates these observations. The decision algorithm compares demand among approaches and determines whether green time should be extended, reduced or retained.

5.2 Working Procedure

7. Collect real-time traffic data from each approach.
8. Estimate traffic density, queue length and waiting demand.
9. Compare demand between approaches.
10. Select a timing adjustment when a meaningful imbalance is detected.
11. Apply the adjustment only within minimum/maximum phase and pedestrian-safety limits.
12. Record the new traffic condition and continue the feedback cycle.

5.3 Example Decision Logic

A prototype can calculate a demand score using queue length, vehicle count and waiting time. Conceptually: Demand Score = $w_1(\text{queue length}) + w_2(\text{vehicle count}) + w_3(\text{waiting time})$. If one approach has a significantly higher score, the controller may extend its green phase within the allowed range. Smoothing or a dead-band should be used so that small measurement fluctuations do not cause continuous signal changes.

6. Feasibility Analysis

6.1 Technical Feasibility

The solution is technically feasible because sensors, cameras, microcontrollers or edge computers, databases and web dashboards are established technologies. A student prototype can use simulated traffic data or a small model intersection. Real-road deployment would require certified traffic controllers, authority approval and professional traffic engineering.

6.2 Economic Feasibility

Prototype costs can be kept low by using simulated inputs and development boards. Real implementation costs depend on sensors, controllers, communications, installation, calibration, cybersecurity and maintenance. Economic feasibility should compare these costs with measurable reductions in delay, fuel use and operational inefficiency.

6.3 Operational Feasibility

Traffic operators should be able to monitor the system, override adaptive operation when necessary and switch to a safe fallback plan during equipment or communication failures. Staff training, calibration and maintenance are therefore important.

6.4 Data and Cybersecurity

The system should protect traffic-control commands and restrict access to authorized users. Data collection should follow data-minimization principles. If cameras are used, unnecessary personal imagery should not be retained; traffic counts or other non-identifying measurements can be processed where appropriate.

6.5 Main Risks and Controls

Risk	Control
Sensor failure	Health checks and fallback timing
Network failure	Local control and safe default
Unstable adaptation	Smoothing and timing limits
Safety conflict	Certified controller constraints
Privacy issue	Data minimization and access control

7. Expected Environmental Benefits

The environmental benefit of adaptive signals comes mainly from reducing avoidable delay and improving traffic flow. When vehicles spend less time waiting or repeatedly starting from long queues, fuel use and emissions may decrease. The actual benefit depends on local traffic, vehicle types, road geometry and signal operation, so site-specific measurement is required rather than assuming a universal percentage reduction.

7.1 Expected Benefits

- Reduced average waiting time at congested intersections.
- Lower unnecessary engine-idling time.
- Potential reduction in fuel consumption caused by avoidable delay.
- Potential reduction in CO₂ and other traffic-related emissions.
- Improved travel-time reliability and traffic flow.
- Better utilization of existing road infrastructure.

7.2 Evaluation Method

The system should be evaluated by comparing a baseline period with a period after adaptive control under similar traffic conditions. Average delay, queue length, number of stops, throughput and travel-time reliability can be measured directly. Emissions can then be estimated using an appropriate fuel-consumption or vehicle-emission model.

Indicator	Comparison
Delay	Before vs. after
Queue length	Before vs. after
Idle time	Estimated before vs. after
Fuel use	Modelled before vs. after
Emissions	Modelled before vs. after

8. Justification of the Proposed Solution

The SATSS is suitable because it directly addresses a major congestion point: the intersection. Instead of only informing travellers about congestion, it improves the operation of the traffic signal itself. One adjustment can benefit multiple vehicles using the intersection. The solution is also measurable, scalable and suitable for a student prototype using simulation or a controlled model.

Its integration with the Sustainable Transportation Planner can provide a dashboard showing traffic density, current signal phase, delay and environmental indicators. This makes the individual solution distinct while allowing it to function as one module within the group's broader sustainable-transportation platform.

9. Conclusion

The Smart Adaptive Traffic Signal System provides a practical engineering and IT solution to traffic congestion and vehicle emissions. By using real-time traffic information to adjust signal timings, the system can allocate green time more effectively, reduce unnecessary waiting and improve the use of existing intersection capacity. The resulting reduction in avoidable idling may also contribute to lower fuel consumption and associated emissions.

The project has a strong sustainability connection, particularly with SDG 11, while also supporting SDG 9 and SDG 13. An ISO 14001-oriented approach can be used to monitor environmental performance and promote continuous improvement. LCA/EIA considerations ensure that the environmental effects of equipment, energy use, maintenance and disposal are also considered.

The system is technically feasible as a student prototype using simulated traffic data, sensors or a small-scale intersection model. For real-world implementation, certified controllers, safety validation, cybersecurity, privacy protection, maintenance and approval from the relevant traffic authority would be essential. Overall, the proposed solution offers a measurable, scalable and sustainability-focused way to reduce intersection waiting and idling within the wider Sustainable Transportation Planner.

10. References

- [1] U.S. Federal Highway Administration (FHWA), Traffic Signal Program Handbook, FHWA-HOP-23-041, 2023.
- [2] United Nations Department of Economic and Social Affairs, Sustainable Development Goal 11: Sustainable Cities and Communities.
- [3] United Nations Department of Economic and Social Affairs, Sustainable Development Goal 9: Industry, Innovation and Infrastructure.
- [4] United Nations Department of Economic and Social Affairs, Sustainable Development Goal 13: Climate Action.
- [5] International Organization for Standardization (ISO), ISO 14001: Environmental management systems — Requirements with guidance for use.
- [6] U.S. Federal Highway Administration, Adaptive Signal Control Technology case-study and guidance materials.